

Joanna X. Ju

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Education

University of California, San Diego

Bachelor of Science in Computer Engineering

Cumulative GPA: 3.70 / 4.00 (UC San Diego & Pasadena City College)

La Jolla, CA

Expected June 2028

Relevant Coursework: Computer Architecture, Digital Circuits, Digital Systems, Systems Programming, Advanced Data Structures, Circuits & Systems, Assembly Language

Projects

Custom 8-bit RISC Processor Design

SystemVerilog, Assembly, C++

- Designed and implemented a custom 8-bit RISC processor with a 9-bit instruction set, 8-register file, ALU, condition flags, branching, and separate instruction and data memories.
- Developed synthesizable SystemVerilog RTL for the datapath, control logic, register file, ALU, and memory interfaces.
- Built a two-pass C++ assembler to translate custom assembly programs into machine code and resolve labels and branch offsets.
- Verified individual modules and full-processor functionality through RTL simulation and synthesis using ModelSim/Questa and Quartus.

Experience

High Performance Computing & Embedded Systems Intern

Sep 2026 – Present

NASA Armstrong Flight Research Center (AFRC)

Edwards, CA

- Support NASA's High-Performance Spaceflight Computing (HPSC) platform and Next-generation Research Flight Controls System (N-RFCS) as part of the high-performance computing and flight control systems team.
- Contribute to the design, development, and optimization of real-time and high-performance software systems in C/C++ for aerospace and embedded systems.
- Work with parallel programming and multithreaded solutions for deterministic, time-sensitive applications.
- Contribute to software development on Unix/Linux systems and real-time operating systems (RTOS), including unit testing and system-level integration.
- Support hardware-in-the-loop (HIL) testing and embedded communication interfaces including UART, CAN, Ethernet, SPI, and I2C.

Software/Avionics Team Member

Oct 2025 – Present

UC San Diego Rocket Propulsion Laboratory (RPL)

La Jolla, CA

- Collaborate with the avionics team to maintain and validate embedded systems for student-built rockets.
- Analyze embedded firmware and hardware interfaces for flight-computer, ground-station, and sensor systems.
- Test system functionality and diagnose hardware-software integration and communication issues.

SPIS Mentor

Aug 2026 – Sep 2026

UC San Diego, Computer Science and Engineering Department

La Jolla, CA

- Mentored incoming computer science and engineering students in algorithmic problem-solving, programming, and debugging.
- Guided student teams in designing, programming, and debugging Raspberry Pi robotics projects using Python.

Computer Science Tutor

Aug 2024 – Jun 2025

Pasadena City College

Pasadena, CA

Math Instructor

Jul 2024 – Aug 2026

Mathnasium

Temple City & San Diego, CA

Skills

Languages: C, C++, Assembly (MIPS, RISC-V, x86), Python, Java, SystemVerilog, Verilog

Architecture & Digital Hardware: Computer Architecture, Processor Design, RISC-V, MIPS, RTL Design, Digital Logic, Digital Systems, Microprocessors, Microcontrollers, Memory Systems, Datapath & Control Design, Instruction Set Architecture (ISA)

Embedded & Systems: Embedded Systems, Bare-Metal Programming, Embedded Linux, RTOS, Microchip PolarFire SoC, Arduino, Raspberry Pi, Hardware-Software Integration

Embedded Linux & Build Systems: Yocto Project, Buildroot

Interfaces: UART, CAN, Ethernet, SPI, I2C

Circuits: Analog Circuit Design, Analog Circuit Analysis, Circuit Analysis, Electronic Circuits

Tools: Linux, Git, GitHub, GitLab, Quartus, ModelSim/Questa, Cadence Virtuoso, PSpice, LTspice, VS Code